Supporting Material: Scale-Bridging Techno-Economic, Microclimatic, and Open-Quantum Assessment of Semi-Transparent Organic Agrivoltaics for Sub-Saharan Perennial Crops

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Abstract

This Supporting Material documents the complete optical transfer-matrix formulation, microscopic 14-site LHCII exciton–vibrational Hamiltonians and MesoHOPS 1.8.0 convergence checks, hourly ERA5 meteorological dataset extraction, tri-crop Farquhar–von Caemmerer–Berry (FvCB)–Medlyn–FAO-56 governing equations, the Penja white pepper (*poivre blanc de Penja*, IGP) versus generic black pepper parameterization, cooperative battery dispatch and life-cycle accounting rules, and supplementary scenario tables.

S1. Optical transfer-matrix formalism and spectral calibration

Multilayer optical propagation across the semi-transparent organic photovoltaic (ST-OPV) stack—glass substrate (1.1 mm, incoherent) / ITO anode (140 nm) / ZnO ETL (30 nm) / PM6:Y6 (1:1.2) photoactive layer ($d_{\mathrm{act}} \in [40, 140]$ nm) / MoO_3 HTL (10 nm) / ultrathin Ag electrode ($d_{\mathrm{Ag}} \in [8, 22]$ nm) / MoO_3 dielectric capping layer ($d_{\mathrm{cap}} \in [0, 60]$ nm)—is evaluated using the 2×2 Abeles characteristic transfer-matrix method at 1 nm spectral resolution ($\lambda \in [300, 1100]$ nm). Unpolarized irradiance responses average transverse-electric (TE) and transverse-magnetic (TM) modes across incident angles $\theta \in [0°, 80°]$:

$$T_{\mathrm{OPV}}^{\mathrm{beam}}(\lambda,\theta) = \frac{1}{2}\left[T_{\mathrm{TE}}(\lambda,\theta) + T_{\mathrm{TM}}(\lambda,\theta)\right], \qquad T_{\mathrm{OPV}}^{\mathrm{diff}}(\lambda) = 2\int_0^{\pi/2} T_{\mathrm{OPV}}^{\mathrm{beam}}(\lambda,\theta)\sin\theta\cos\theta\,\mathrm{d}\theta. \quad \text{(S1)}$$

Complex refractive indices $n(\lambda) + i\kappa(\lambda)$ use published variable-angle spectroscopic ellipsometry datasets for PM6:Y6, D18:Y6, and ITO alongside dispersive Johnson–Christy Ag optical constants, avoiding constant-index artifacts. Across all 120 Pareto-swept stacks, local energy conservation satisfies $\max|R(\lambda) + T(\lambda) + A_{\mathrm{tot}}(\lambda) - 1| = 4.44 \times 10^{-16}$. An opaque reference cell ($d_{\mathrm{act}} = 75$ nm, $d_{\mathrm{Ag}} = 100$ nm) reproduces the experimental PM6:Y6 benchmark ($J_{\mathrm{sc}} = 25.701\,\mathrm{mA\,cm^{-2}}$, PCE = 15.501 % vs. $26.36\,\mathrm{mA\,cm^{-2}}$ and 16.27 % reported by Shen et al. [1]).

Photosynthetically active transmittance (APT, 400 to 700 nm), extended PAR (ePAR, 400 to 750 nm), light utilization efficiency (LUE = PCE × APT), and photomorphogenic red-to-far-red shift (Δ(R:FR) comparing [655, 665] nm to [725, 735] nm) follow AM1.5G photon-flux weighting (ASTM G-173-03). The 24-design Pareto frontier yields $\mathrm{LUE}_{\max} = 4.424$ % (APT $\in$ [14.54 %, 52.06 %]), with the focal stack APT44 ($d_{\mathrm{act}} = 60$ nm, $d_{\mathrm{Ag}} = 12$ nm, $d_{\mathrm{cap}} = 35$ nm) delivering APT = 44.12 % and PCE = 9.93 %.

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S2. Microscopic 14-site LHCII exciton–vibrational model and CW NESS bounds

S2.1. Three independent 14-site Hamiltonians and dipoles

To eliminate single-Hamiltonian bias, three monomeric LHCII 14-pigment Frenkel-exciton Hamiltonians (sites Chl b_{601}–a_{614}) are constructed directly from primary literature: H1 takes the Redfield–Förster pigment-site energies and couplings of Novoderezhkin et al. [2] paired with point-dipole transition geometry from Liu et al. (PDB 1RWT); H2 adopts the electrostatic charge-density coupling site energies and Poisson–TrEsp couplings of Müh et al. [3] (PDB 2BHW); and H3 implements the MMSIC-MD parameter set of Zhu, Higashi, and Saito [4] and Kim et al. [5], with a symmetric Hermitian coupling matrix ($\max|H_{ij}-H_{ji}| = 0.0\,\mathrm{cm}^{-1}$) led by Chl a_{611}–a_{612} at $+129.0\,\mathrm{cm}^{-1}$ and Chl a_{604}–b_{606} at $+114.0\,\mathrm{cm}^{-1}$. Transition dipole strengths ($\mu^2_{\mathrm{Chl}\,a} = 21.0\,\mathrm{D}^2$, $\mu^2_{\mathrm{Chl}\,b} = 14.7\,\mathrm{D}^2$) preserve exact non-orthogonality along the crystallographic N_B–N_D molecular axes (dimer 610–611 in PDB 1RWT: $\cos\theta = +0.932$, $\kappa = -1.812$), enabling partial-secular Fano interference testing. Each chlorophyll couples to a Drude–Lorentz overdamped protein bath ($\lambda_D = 50.87\,\mathrm{cm}^{-1}$ matching the Renger spectral density integral [5], bounded over $[35.0, 140.0]\,\mathrm{cm}^{-1}$, $\gamma_D = 50.0\,\mathrm{cm}^{-1}$) plus high-frequency intramolecular modes (340 and $740\,\mathrm{cm}^{-1}$) at $T = 295\,\mathrm{K}$ across $N = 200$ static Gaussian disorder realizations (FWHM $= 110\,\mathrm{cm}^{-1}$).

S2.2. Continuous-wave non-equilibrium steady state vs. pulsed MesoHOPS

Under continuous-wave (CW) solar illumination filtered by $T_{\mathrm{OPV}}(\lambda)$, broadband photon-pumping operators act without inter-mode temporal phase correlation [6, 7]. Solving the Liouville steady state $\mathcal{L}\rho_{\mathrm{ss}} = 0$ compares the full Bloch–Redfield partial-secular tensor (η_{PS}) against its diagonal secular Pauli counterpart (η_{Pauli}) via

$$\Delta\eta_{\mathrm{NESS}} = \eta_{\mathrm{PS}} - \eta_{\mathrm{Pauli}}, \qquad \eta = \frac{k_{\mathrm{RC}}\,\mathrm{Tr}(P_{\mathrm{term}}\rho_{\mathrm{ss}})}{k_{\mathrm{RC}}\,\mathrm{Tr}(P_{\mathrm{term}}\rho_{\mathrm{ss}}) + k_{\mathrm{NR}}\,\mathrm{Tr}(\rho_{\mathrm{ss}}) + k_{\mathrm{NPQ}}\,\mathrm{Tr}(P_{\mathrm{qE}}\rho_{\mathrm{ss}})}, \tag{S2}$$

with open-RC trapping $k_{\mathrm{RC}} = 0.005\,\mathrm{ps}^{-1}$ (swept over $[0.0033, 0.5]\,\mathrm{ps}^{-1}$), nonradiative decay $k_{\mathrm{NR}} = 0.25\,\mathrm{ns}^{-1}$, and q_E photoprotective quenching $k_{\mathrm{NPQ}} \in \{0, 1.5\}\,\mathrm{ns}^{-1}$. A 4-site exact QuTiP-BoFiN HEOM cross-check [8] verifies $|\eta_{\mathrm{HEOM}} - \eta_{\mathrm{Pauli}}| = 0.002\,10$ (H2) and $0.000\,106$ (H3).

Separately, full non-Markovian 14-site MesoHOPS 1.8.0 trajectories ($N = 48$ disorder seeds, 2000 fs propagation, max trace-norm deviation $\leq 1.11 \times 10^{-15}$) confirm that coherent transient wavepacket dynamics under pulsed initial conditions deviate markedly from secular kinetics (terminal Chl a_{610}–a_{612} MAE = 0.0198 on H2–DL, 0.2017 on H3–DL, and 0.2200 on H3–DL+Vib$_{340}$, where $340\,\mathrm{cm}^{-1}$ vibronic dynamics reach $p_{\mathrm{term}} = 0.0835$ vs. 0.4035 Pauli), yet vanish under CW incoherent sunlight ($|\Delta\eta_{\mathrm{NESS}}| \leq 3.370 \times 10^{-5}$).

S3. ERA5 microclimate, tri-crop ecophysiology, and Penja white pepper parameterization

Hourly meteorological drivers (8760 h, 2023 UTC) are extracted from Open-Meteo ERA5 reanalysis [9] for Foumbot (1044 m, Arabica coffee), Njombe-Penja (273 m, Penja white pepper), and Djuttitsa (1400 m, highland tea). Diffuse fractions are obtained from ERA5 direct/diffuse decomposition (annual averages 0.365, 0.461, and 0.412; wet season 0.416 to 0.502; dry season 0.260 to 0.360). Sub-canopy radiation solves 2D row-geometry shadow projection with diffuse sky view factor F_{sky} (generating low-PPFD shading fractions $F_{\mathrm{lp}} \in [0.277, 0.544]$ under ST-OPV depending on GCR). Coupled leaf and module temperatures resolve hourly sensible, radiative, and latent energy balances (residual $\leq 1.569 \times 10^{-5}\,\mathrm{W\,m}^{-2}$). Crop assimilation couples the Farquhar–von Caemmerer–Berry (FvCB) model [10] with Medlyn optimal stomatal conductance [11] and daily FAO-56 two-bucket water stress (AWC = 160 mm) [12],

plus calibrated irradiance photoinhibition thresholds ($I_{\mathrm{PI}} = 900\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ for *Piper nigrum*, $1200\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ for *Camellia sinensis*). Exact Shapley causal decomposition across all $2^4 = 16$ coalition subsets guarantees machine-precision additivity (residual $\leq 7.105 \times 10^{-15}\,\%$).

S3.1. Penja white pepper (poivre blanc de Penja) vs. black pepper parameterization

Table S1 documents why Penja white pepper parameters avoid conflation with whole-dried black pepper.

Table S1: Agronomic, post-harvest processing, and market parameterization separating Protected Geographical Indication (PGI) Penja white pepper from generic black pepper.

Parameter	**Model Value (White)**	**Black Pepper Contrast**	**Agronomic / Market Basis**
Clean dry yield	$800\,\mathrm{kg\,ha^{-1}\,yr^{-1}}$	$1100\,\mathrm{kg\,ha^{-1}\,yr^{-1}}$ equivalent	White kernel recovery is ∼24 % of red-ripe fresh berries vs. ∼34 % whole-dried black pepper.
Baseline farm-gate price	$5000\,\mathrm{XAF\,kg^{-1}}$	1800 to $2800\,\mathrm{XAF\,kg^{-1}}$	Cameroon Njombe-Penja peeled white kernel price (3500 to $6000\,\mathrm{XAF\,kg^{-1}}$ range); PGI export tier $= 14\,000\,\mathrm{XAF\,kg^{-1}}$.
Post-harvest electric load	$45.0\,\mathrm{kWh\,t^{-1}}$ (85 % day)	10 to $15\,\mathrm{kWh\,t^{-1}}$ (sun drying)	Sized for spring-water retting circulation pumps (7 to 10 d), mechanical decortication/washing, and fan-assisted solar drying.
Gas-exchange ($V_{c\max 25}, J_{\max 25}$)	44.6, $65.0\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$	Species-shared (*Piper nigrum* L.)	Calibrated in steady-state FvCB against Oliveira et al. [13] and Ferreira et al. [14].

S4. Extended scenario results (S_0–S_5)

Table S2: Summary of focal agronomic, PV generation, and techno-economic outputs under grid-connected operation at the baseline $0\,\mathrm{USD\,tCO_2e^{-1}}$ carbon price and 0 % CAPEX subsidy. NPV and LCA use a 2 h LFP buffer for coffee/pepper and overnight withering BESS for tea in S_3(APT44, GCR65), S_5(APT44, GCR65, irrig.), and S_2(c-Si, GCR40), whereas S_3(APT56, GCR50) and S_3(APT52, GCR65) reflect daytime-only dispatch without storage.

Crop	**Scenario**	**Shade** (%)	ΔY_{ann} (%)	ΔY_{dry} (%)	ΔY_{wet} (%)	**PV Yield** ($\mathrm{MWh\,ha^{-1}\,yr^{-1}}$)	**LER** (–)	**NPV** ($\mathrm{kUSD\,ha^{-1}}$)	**LCA** ($\mathrm{gCO_2e\,kWh^{-1}}$)
Coffee	S3 (APT56, GCR50)	25.7	−1.21	+9.02	−6.96	717.0	1.265	213.0	33.99
Coffee	S3 (APT44, GCR65)	40.6	−7.08	+8.82	−15.88	1223.3	1.401	403.5	26.16
Coffee	S5 (APT44, GCR65, irrig.)	40.6	+0.22	+38.23	−15.88	1223.3	1.474	414.2	26.18
Coffee	S2 (c-Si, GCR40)	40.0	−6.38	+8.73	−14.92	1479.0	1.507	779.3	18.92
Penja Pepper	S3 (APT56, GCR50)	25.9	+7.27	+19.57	+1.53	611.7	1.350	205.7	39.84
Penja Pepper	S3 (APT52, GCR65)	36.0	+8.16	+27.56	+0.04	870.2	1.477	253.3	36.85
Penja Pepper	S3 (APT44, GCR65)	40.8	+7.43	+30.24	−1.36	1043.7	1.548	368.4	30.73
Penja Pepper	S5 (APT44, GCR65, irrig.)	40.8	+12.25	+49.68	−1.36	1043.7	1.596	376.5	30.74
Penja Pepper	S2 (c-Si, GCR40)	40.0	+6.95	+27.89	−0.78	1257.2	1.640	709.1	22.31
Highland Tea	S3 (APT56, GCR50)	25.8	+0.01	+11.26	−3.47	652.5	1.271	556.3	37.35
Highland Tea	S3 (APT44, GCR65)	40.7	−3.35	+15.98	−8.84	1113.3	1.429	860.8	32.66
Highland Tea	S5 (APT44, GCR65, irrig.)	40.7	+2.47	+41.38	−8.84	1113.3	1.487	899.8	32.91
Highland Tea	S2 (c-Si, GCR40)	40.0	−2.30	+16.56	−7.65	1374.4	1.548	1272.6	23.46

S4.1. Sobol robustness checks across exact-HEOM quantum bounds and zero-carbon pricing

To test sensitivity under stricter counterfactual bounds, all $N = 10\,000$ Saltelli sequences (140 000 model evaluations per crop under an identical quasi-Monte Carlo sequence) are re-evaluated across four uncertainty-set variants (Table S3): (1) the baseline policy ensemble ($\Delta\eta_{\mathrm{NESS}} \in [-0.02, +0.10]\,\%$, carbon price $[0, 25]\,\mathrm{USD\,tCO_2e^{-1}}$); (2) the exact-HEOM quantum envelope ($\Delta\eta_{\mathrm{NESS}} \in [-0.21, +0.21]\,\%$, matching $|\eta_{\mathrm{HEOM}} - \eta_{\mathrm{Pauli}}| = 2.10 \times 10^{-3}$); (3) strict zero-carbon Monte Carlo ($0\,\mathrm{USD\,tCO_2e^{-1}}$); and (4) the combined exact-HEOM and zero-carbon configuration. Even under the widest combined envelope, comparative quantum total-order Sobol sensitivity never exceeds 2.59×10^{-7} on ΔNPV or 9.01×10^{-8} on standalone $\mathrm{NPV_{OPV}}$, while exceedance odds remain bounded within 11.0 to 24.7 %. In contrast, classical macro-financial parameters govern standalone $\mathrm{NPV_{OPV}}$ (grid export tariff $S_{Ti} = 0.5176$ in coffee and 0.4263 in pepper; WACC $S_{Ti} = 0.4503$ and diesel price $S_{Ti} = 0.2817$ in highland tea), whereas hardware parameters ($C_{\mathrm{mod}}^{\mathrm{OPV}}$, T_{80}, and $C_{\mathrm{BOS}}^{\mathrm{c\text{-}Si}}$) govern comparative parity ΔNPV as summarized in Main Text Table 5.

Table S3: Robustness of Saltelli–Sobol exceedance probabilities $P(\mathrm{NPV_{OPV}} > \mathrm{NPV_{c\text{-}Si}})$ and quantum total-order sensitivity $S_{Ti}(\Delta\eta_{\mathrm{NESS}})$ evaluated on ΔNPV across $N = 10\,000$ base samples.

Variant	**Crop**	$P(>$ c-Si) (%)	$\Delta\mathrm{NPV}_{p50}$ ($\mathrm{kUSD\,ha^{-1}}$)	$S_{Ti}^{\mathrm{Q}}(\Delta\mathrm{NPV})$ (–)	$S_{Ti}^{\mathrm{Q}}(\mathrm{NPV_{OPV}})$ (–)	$S_{Ti}(C_{\mathrm{mod}}^{\mathrm{OPV}})$ (–)
Baseline	Coffee	16.75	−139.3	3.83×10^{-9}	1.16×10^{-9}	0.343
Baseline	Penja Pepper	22.99	−105.3	2.10×10^{-8}	7.15×10^{-9}	0.366
Baseline	Highland Tea	11.00	−176.0	9.85×10^{-9}	1.89×10^{-9}	0.323
HEOM (±0.21 %)	Coffee	16.74	−139.3	4.69×10^{-8}	1.42×10^{-8}	0.343
HEOM (±0.21 %)	Penja Pepper	22.97	−105.3	2.58×10^{-7}	8.76×10^{-8}	0.366
HEOM (±0.21 %)	Highland Tea	11.00	−176.0	1.21×10^{-7}	2.32×10^{-8}	0.323
Zero-Carbon	Coffee	18.51	−131.0	3.87×10^{-9}	1.20×10^{-9}	0.346
Zero-Carbon	Penja Pepper	24.71	−97.9	2.12×10^{-8}	7.35×10^{-9}	0.369
Zero-Carbon	Highland Tea	12.34	−166.5	9.95×10^{-9}	1.95×10^{-9}	0.327
HEOM + Zero-Carbon	Coffee	18.51	−131.0	4.74×10^{-8}	1.46×10^{-8}	0.346
HEOM + Zero-Carbon	Penja Pepper	24.72	−97.9	2.59×10^{-7}	9.01×10^{-8}	0.369
HEOM + Zero-Carbon	Highland Tea	12.33	−166.6	1.22×10^{-7}	2.39×10^{-8}	0.327

References

[1] Z. Shen, G. Lu, X. Tang, et al., Intrinsic light reflection of conjugated films towards high-performance organic solar cells, Nat. Commun. 17 (2026) 655. doi:10.1038/s41467-025-67379-z.

[2] V. I. Novoderezhkin, A. Marin, R. van Grondelle, Intra- and inter-monomeric transfers in the light harvesting LHCII complex: the Redfield-Förster picture, Phys. Chem. Chem. Phys. 13 (38) (2011) 17093–17103. doi:10.1039/c1cp21079c.

[3] F. Müh, M. E.-A. Madjet, T. Renger, Structure-based identification of energy sinks in plant light-harvesting complex II, J. Phys. Chem. B 114 (42) (2010) 13517–13535. doi:10.1021/jp106323e.

[4] Z. Zhu, M. Higashi, S. Saito, Development of molecular dynamics parameters and theoretical analysis of excitonic and optical properties in the light-harvesting complex II, J. Chem. Theory Comput. 21 (2025) 413–427. doi:10.1021/acs.jctc.4c01214.

[5] E. Kim, D. Lee, S. Sakamoto, J.-Y. Jo, M. Vargas, A. Ishizaki, et al., Network analysis with quantum dynamics clarifies why photosystem II exploits both chlorophyll a and b, Sci. Adv. 11 (2025) eads0327. doi:10.1126/sciadv.ads0327.

[6] P. Brumer, M. Shapiro, Molecular response in one-photon absorption via natural thermal light vs. pulsed laser excitation, Proc. Natl. Acad. Sci. U.S.A. 109 (48) (2012) 19575–19578. doi:10.1073/pnas.1211209109.

[7] I. Kassal, J. Yuen-Zhou, S. Rahimi-Keshari, Does coherence enhance transport in photosynthesis?, J. Phys. Chem. Lett. 4 (3) (2013) 362–367. doi:10.1021/jz301872b.

[8] N. Lambert, T. Raheja, S. Cross, P. Menczel, S. Ahmed, A. Pitchford, D. Burgarth, F. Nori, Qutip-bofin: A bosonic and fermionic numerical hierarchical-equations-of-motion library with applications in light-harvesting, quantum control, and single-molecule electronics, Physical Review Research 5 (1) (2023) 013181. doi:10.1103/physrevresearch.5.013181.

[9] H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, et al., The ERA5 global reanalysis, Q. J. R. Meteorol. Soc. 146 (730) (2020) 1999–2049. doi:10.1002/qj.3803.

[10] G. D. Farquhar, S. von Caemmerer, J. A. Berry, A biochemical model of photosynthetic CO_2 assimilation in leaves of C_3 species, Planta 149 (1) (1980) 78–90. doi:10.1007/BF00386231.

[11] B. E. Medlyn, R. A. Duursma, D. Eamus, D. S. Ellsworth, I. C. Prentice, C. V. M. Barton, K. Y. Crous, P. de Angelis, M. Freeman, L. Wingate, Reconciling the optimal and empirical approaches to modelling stomatal conductance, Global Change Biology 17 (6) (2011) 2134–2144. doi:10.1111/j.1365-2486.2010.02375.x.

[12] R. G. Allen, L. S. Pereira, D. Raes, M. Smith, Crop evapotranspiration: Guidelines for computing crop water requirements, FAO Irrigation and Drainage Paper 56, Food and Agriculture Organization of the United Nations, Rome (1998).

[13] M. G. Oliveira, G. Oliosi, F. L. Partelli, J. C. Ramalho, Physiological responses of photosynthesis in black pepper plants under different shade levels promoted by intercropping with rubber trees, Ciênc. Agrotecnol. 42 (5) (2018) 513–526. doi:10.1590/1413-70542018425020418.

[14] T. R. Ferreira, V. P. Sallin, B. C. Neto, J. Crasque, A. Pires, P. S. Rodrigues, et al., Morphophysiological responses of black pepper to recurrent water deficit, Photosynthetica 62 (2024) 292–301. doi:10.32615/ps.2024.030.

S5. Coupled structural ($AWC \times g_b$), meteorological stress, and seasonal monsoon-open louver audits

To quantify biophysical parameter uncertainty that enters upstream of the 12-parameter Sobol QMC ensemble (Section 5.3), `scripts/wp3b_structural_and_louver_audit.py` propagates the full 8760 h coupled FvCB–Medlyn–FAO-56–LWIR engine across two non-Sobol axes without altering baseline canonical datasets: (i) a coupled 3×3 factorial sweep of rainfed root-zone available water capacity AWC $\in \{120, 160, 220\}$ mm and leaf boundary-layer conductance $g_b \in \{0.6, 1.2, 2.5\}$ mol m^{-2} s^{-1}, and (ii) two physically consistent synthetic meteorological stress perturbations around the verified 2023 ERA5 forcing ("warm–dry": $\Delta T_{\text{air}} = +1.5\,^{\circ}$C, ΔRH $= -5$ pp, precipitation $\times 0.75$, GHI $\times 1.05$, $\Delta f_{\text{diff}} = -0.05$; "cool–cloudy": $\Delta T_{\text{air}} = -1.0\,^{\circ}$C, ΔRH $= +5$ pp, precipitation $\times 1.25$, GHI $\times 0.90$, $\Delta f_{\text{diff}} = +0.08$, recomputing e_a and ambient VPD via Tetens saturation). Each corner is referenced to an open-sun control (S_0)

evaluated under identical (AWC, g_b) or climate forcing (Table S4). Across all three crops, wind-coupled boundary-layer conductance g_b dominates soil AWC by more than an order of magnitude: dense, calmer canopies ($g_b = 0.6\,\mathrm{mol\,m^{-2}\,s^{-1}}$) elevate open-sun leaf temperature and render agrivoltaic cooling strongly yield-positive (+5.20 to +29.17 %), whereas well-mixed windy canopies ($g_b = 2.5\,\mathrm{mol\,m^{-2}\,s^{-1}}$) keep open-sun leaves near ambient air temperature and shift shaded yields mildly light-limited (−13.40 to −1.88 %), while preserving narrow inter-technology parity (−1.14 to +2.41 pp).

We also quantify the counterfactual seasonal retraction design (S_6, Table S5) suggested qualitatively for cloudy wet months in Section 5.1, wherein panels on S_3 (PM6:Y6 ST-OPV, GCR = 0.65) and S_2 (c-Si, GCR = 0.40) retract edge-on to full open sky from June through October ($\mathrm{GCR}_h = 0$, $\tau_{\mathrm{PAR},h} = 1$) while remaining closed November–May. Re-evaluating the coupled 8760 h crop engine, factory/BESS dispatch, 20 yr DCF economics, and dual-FU ISO 14040 LCA shows that although June–October retraction rescues monsoon photosynthesis by +1.22 to +7.38 pp (S_3) and +0.98 to +6.98 pp (S_2), the associated 38.0 to 40.1 % loss in annual electricity revenue reduces 20 yr NPV by 349 596 to 784 209 USD ha^{-1} relative to fixed arrays (and by 357 696 to 826 855 USD ha^{-1} relative to S_5 dry-season drip irrigation), yielding zero (0.00 USD m^{-2}) capital headroom for motorized louver actuation wherever grid-connected power monetization anchors bankability.

Table S4: Coupled 3×3 (AWC × g_b) structural sensitivity and synthetic 2023 meteorological stress perturbations for PM6:Y6 ST-OPV (S_3, APT = 44 %, GCR = 0.65) versus opaque c-Si (S_2, GCR = 0.40). Each yield shift ΔY is referenced to open sun (S_0) under the same parameter or climate corner; $S_3 - S_2$ denotes percentage-point contrast (pp).

Condition / corner	AWC	g_b	Coffee (ΔY, %)			Pepper / Tea (ΔY, %)		
	(mm)	(mol m^{-2} s^{-1})	S_3	S_2	$S_3 - S_2$	S_3 (Pep / Tea)	S_2 (Pep / Tea)	$S_3 - S_2$
Low AWC, sheltered	120	0.6	+5.73	+5.97	−0.24	+25.86 / +5.20	+23.57 / +6.15	+2.29 / −0.95
Low AWC, baseline g_b	120	1.2	−7.25	−6.54	−0.71	+6.24 / −3.62	+5.78 / −2.56	+0.47 / −1.06
Low AWC, ventilated	120	2.5	−13.40	−12.44	−0.96	−4.42 / −8.21	−3.95 / −7.07	−0.47 / −1.14
Baseline AWC, sheltered	160	0.6	+5.97	+6.20	−0.23	+27.34 / +5.55	+25.00 / +6.50	+2.34 / −0.94
Canonical 2023 baseline	160	1.2	−7.08	−6.38	−0.70	+7.43 / −3.35	+6.95 / −2.30	+0.48 / −1.05
Baseline AWC, ventilated	160	2.5	−13.26	−12.30	−0.96	−3.46 / −7.98	−2.98 / −6.85	−0.47 / −1.13
High AWC, sheltered	220	0.6	+6.42	+6.63	−0.22	+29.17 / +6.13	+26.76 / +7.02	+2.41 / −0.89
High AWC, baseline g_b	220	1.2	−6.78	−6.09	−0.69	+8.87 / −2.90	+8.37 / −1.90	+0.50 / −1.00
High AWC, ventilated	220	2.5	−13.01	−12.06	−0.95	−2.34 / −7.61	−1.88 / −6.52	−0.47 / −1.09
Synthetic warm–dry stress	160	1.2	+1.28	+1.72	−0.44	+16.67 / +1.67	+15.29 / +2.65	+1.38 / −0.98
Synthetic cool–cloudy stress	160	1.2	−12.82	−11.91	−0.91	+0.74 / −7.77	+0.79 / −6.66	−0.05 / −1.10

Table S5: Quantified counterfactual performance of seasonal monsoon-open louvers (S_6, retracted June–October to full sky) relative to fixed rainfed arrays (S_3/S_2) and dry-season irrigated fixed arrays (S_5) under baseline 20 yr zero-subsidy economics.

Crop	Array	ΔY (%)		Rescue	PV (MWh ha^{-1} yr^{-1})		LCOE (USD kWh^{-1}) / CO_2e (gCO_2e kWh^{-1})		$\Delta\mathrm{NPV}_{S_6-\mathrm{fix}}$
		Fixed	S_6	(pp)	Fixed	S_6	Fixed	S_6	(USD ha^{-1})
Coffee	ST-OPV (S_3)	−7.08	+0.30	+7.38	1223.3	738.8	0.0637 / 26.2	0.1056 / 43.3	−379 337
Coffee	c-Si (S_2)	−6.38	+0.60	+6.98	1479.0	890.8	0.0380 / 18.9	0.0631 / 31.4	−468 579
Pepper	ST-OPV (S_3)	+7.43	+8.65	+1.22	1043.7	627.3	0.0750 / 30.7	0.1248 / 51.1	−349 596
Pepper	c-Si (S_2)	+6.95	+7.93	+0.98	1257.2	752.8	0.0449 / 22.3	0.0750 / 37.3	−425 667
Tea	ST-OPV (S_3)	−3.35	+1.50	+4.85	1113.3	689.8	0.0886 / 32.7	0.1445 / 53.0	−678 246
Tea	c-Si (S_2)	−2.30	+1.97	+4.26	1374.4	849.2	0.0555 / 23.5	0.0909 / 38.2	−784 209